



Parallel & Distributed Computing

Lecture 4:

Flynn's Taxonomy

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What is Flynn's taxonomy

- Flynn's taxonomy is a categorization of forms of parallel computer architectures.
- From the viewpoint of the assembly language programmer, parallel computers are classified by the concurrency in processing sequences (or streams), data, and instructions.

Instruction Streams

one

many

Data Streams

one

many

SISD

traditional von
Neumann single
CPU computer

MISD

May be pipelined
Computers

SIMD

Vector processors
fine grained data
Parallel computers

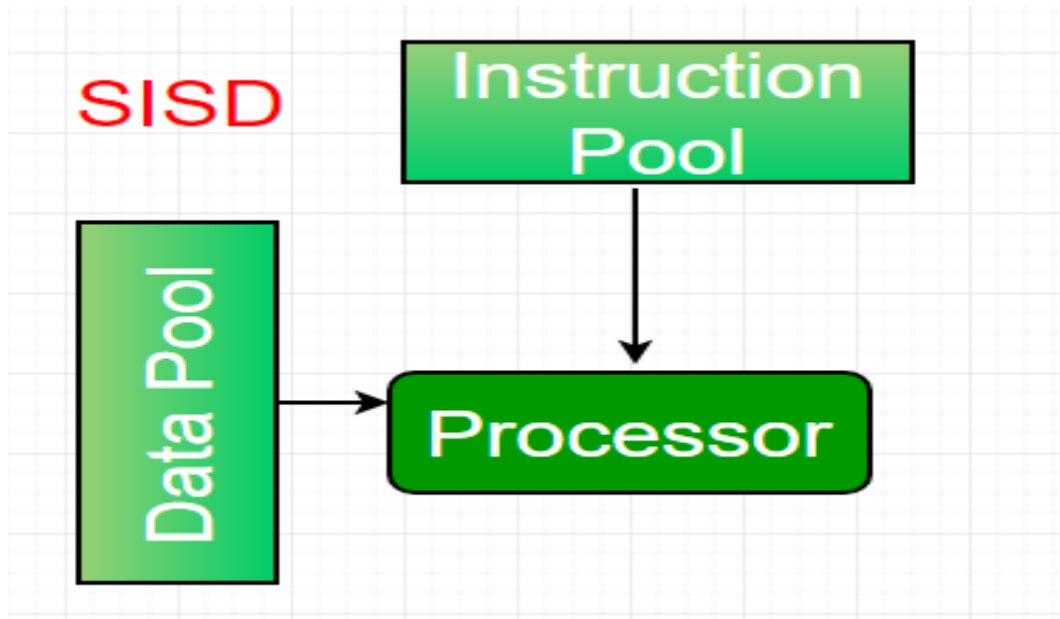
MIMD

Multi computers
Multiprocessors

Single-instruction, single-data (SISD)

- **Single-instruction, single-data (SISD) systems –**
 - An SISD computing system is a uniprocessor machine capable of executing a single instruction, operating on a single data stream.
 - In SISD, machine instructions are processed sequentially and computers adopting this model are popularly called sequential computers.
 - Most conventional computers have SISD architecture. All the instructions and data to be processed must be stored in primary memory.

Single-instruction, single-data

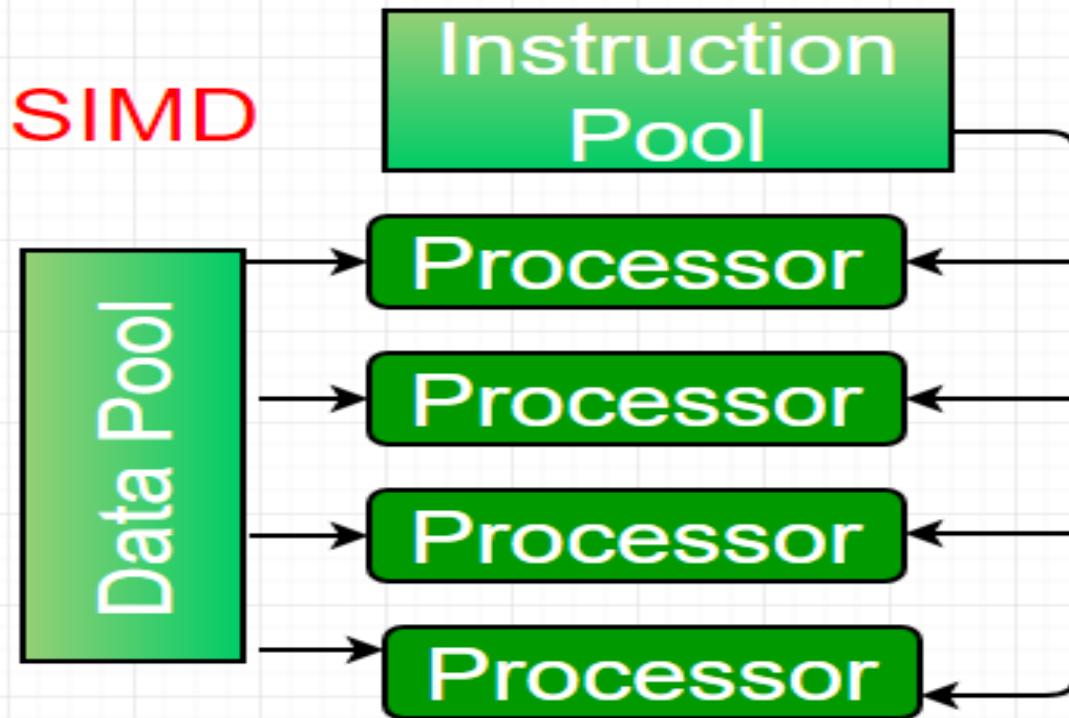


Dominant representative SISD systems are IBM PC, and workstations.

Single-instruction, multiple-data (SIMD)

- **Single-instruction, multiple-data (SIMD) systems –**
 - An SIMD system is a multiprocessor machine capable of executing the same instruction on all the CPUs but operating on different data streams.
 - Machines based on an SIMD model are well suited to scientific computing since they involve lots of vector and matrix operations.
 - So that the information can be passed to all the processing elements (PEs) organized data elements of vectors can be divided into multiple sets(N-sets for N PE systems) and each PE can process one data set.

Single-instruction, multiple-data (SIMD)



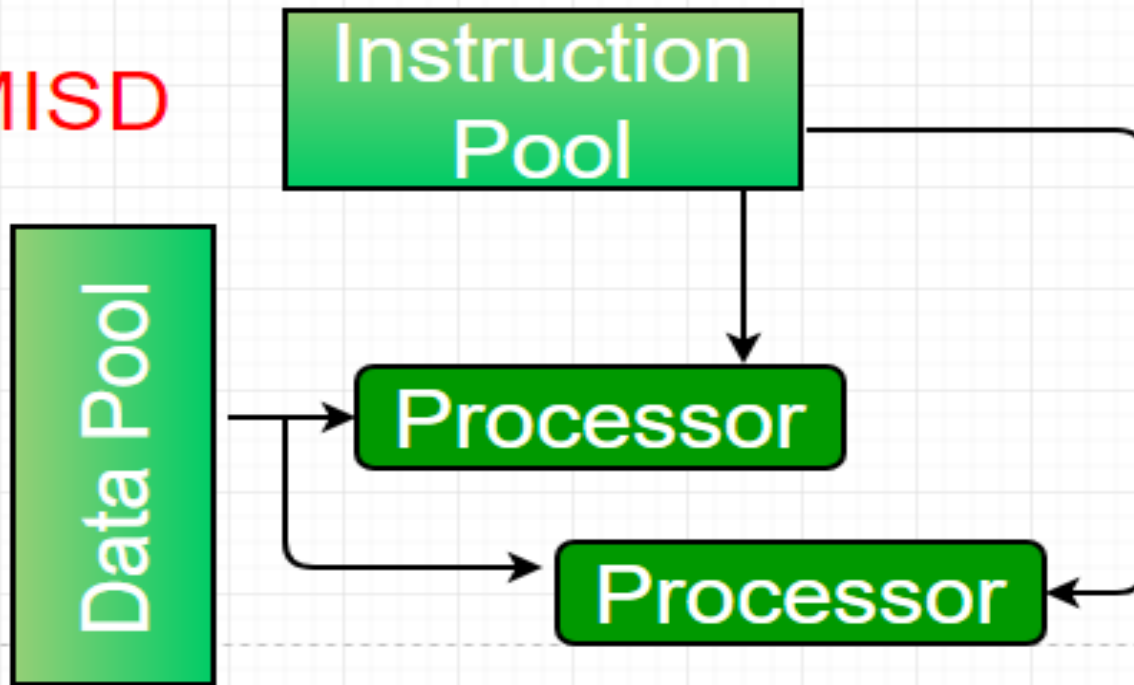
Dominant representative SIMD systems is Cray's vector processing machine.

Multiple-instruction, single-data (MISD)

- Multiple-instruction, single-data (MISD) systems
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 - An MISD computing system is a multiprocessor machine capable of executing different instructions on different PEs but all of them operating on the same dataset

Multiple-instruction, single-data (MISD)

MISD



Multiple-instruction, multiple-data (MIMD)

- **Multiple-instruction, multiple-data (MIMD) systems –**
 - An MIMD system is a multiprocessor machine which is capable of executing multiple instructions on multiple data sets.
 - Each PE in the MIMD model has separate instruction and data streams; therefore machines built using this model are capable to any kind of application.
 - Unlike SIMD and MISD machines, PEs in MIMD machines work asynchronously.

Multiple-instruction, multiple-data (MIMD)

